MACHINE LEARNING METHODOLOGIES V2

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Why Feature Selection Is Necessary

A machine learning model may not always perform well when all available features are included. Some variables may contribute little or nothing to the prediction, while others may introduce unnecessary complexity.

Feature selection helps address this by identifying a smaller group of useful features that can be supplied to the model. Using a more relevant feature set can make the model easier to train and may improve its predictive performance, although better accuracy is not guaranteed in every situation.

Meaning of Feature Selection

Feature selection is the process of identifying the most useful variables for a machine learning task while eliminating features that are irrelevant or provide limited value.

The objective is not simply to reduce the number of columns, but to retain the variables that contain meaningful information for the model.

Importance of Feature Selection

Feature selection is useful because it:

- Helps reduce the **Curse of Dimensionality**.
- Lowers the computational resources required during model development.
- Makes it easier for the learning algorithm to identify meaningful patterns.
- Can contribute to better model performance.

Feature Selection Statistics

The statistical method used for selecting features depends on the nature of the variables involved.

Both the **input features** and the **target/output variable** may be categorical or numerical. Therefore, the type of input and output variables should be considered when deciding which statistical technique to apply.

Major Feature Selection Approaches

Feature selection techniques can generally be divided into:

- **Supervised Feature Selection**

- **Unsupervised Feature Selection**

Supervised Feature Selection

Supervised feature selection makes use of the **target/output variable** when determining which features are useful.

Common supervised approaches include:

- Filter Methods
- Wrapper Methods
- Embedded Methods

Unsupervised Feature Selection

Unsupervised feature selection does **not require a target/output variable**. The selection process is based on characteristics within the input data itself.

Filter Methods

Filter methods evaluate features using statistical or mathematical measures without depending directly on a particular machine learning model.

Common techniques include:

Information Gain

Information Gain measures how much uncertainty is reduced when a feature is used. It is commonly associated with the reduction of **entropy**.

Chi-Squared Test

The Chi-Squared test can be used to examine the relationship or association between categorical variables.

Fisher's Score

Fisher's Score evaluates the usefulness of features and can be used to rank them, with stronger features receiving higher positions in the ranking.

Missing Value Ratio

This method examines how much missing information exists within each feature and can remove features whose missing-value proportion exceeds a specified threshold.

Advantages of Filter Methods

- Generally have lower computational complexity.
- Do not depend on a particular classifier.
- Are relatively fast and scalable, especially for larger datasets.

Limitations of Filter Methods

- They may provide less precise feature selection in some situations.
- They may overlook relationships or dependencies between features.
- They do not directly consider how a particular machine learning model will use the selected features.

Wrapper Methods

Wrapper methods evaluate different combinations or subsets of features by actually using a machine learning model. The selected features are therefore closely connected to the performance of the chosen model.

Forward Selection

Forward Selection starts with an empty feature set. Features are gradually added, and after each addition, the model is evaluated to determine whether the new feature improves the result.

Backward Selection

Backward Selection begins with all available features. Features are progressively removed, usually starting with the least useful ones, while the model is repeatedly evaluated.

Exhaustive Feature Selection

Exhaustive Feature Selection evaluates feature subsets using a brute-force approach. It attempts to examine a large number of possible combinations to identify a suitable feature set.

Recursive Feature Elimination (RFE)

RFE is a recursive, greedy approach. It repeatedly trains a model, determines the importance of the features, removes the less important ones, and continues the process until the desired number of features remains.

Advantages of Wrapper Methods

- They directly interact with the machine learning model.
- The selected features are evaluated according to model performance.
- They can identify feature subsets that work particularly well with a specific model.

Limitations of Wrapper Methods

- They can require considerable computational resources.
- They can take longer to execute, especially with many features.
- There is a greater possibility of overfitting if the selection process is not properly controlled.

Embedded Methods

Embedded methods perform feature selection as part of the model-training process. Instead of treating feature selection as a completely separate step, the learning algorithm identifies useful features while building the model.

Examples include:

- **L1 Regularization (Lasso)**
- **L2 Regularization (Ridge)**
- **Elastic Net**
- **Decision Tree-based methods**
- **Random Forest-based methods**

Embedded methods provide a middle ground between filter and wrapper approaches because feature selection is influenced by the learning algorithm without requiring the same level of repeated model evaluation associated with many wrapper techniques.

Comparing the Main Methods

Filter Method

Uses statistical characteristics of the data to assess features.

Main strength: Fast, scalable, and independent of a specific model.

Main limitation: Does not directly consider feature interactions with the final model.

Wrapper Method

Uses a machine learning model to evaluate different feature subsets.

Main strength: Feature selection is directly connected to model performance.

Main limitation: Can be computationally expensive and may increase the risk of overfitting.

Embedded Method

Selects features during the model-training process.

Main strength: Provides a balance between computational efficiency and model-specific feature selection.

Main limitation: The selection process is generally tied to the learning algorithm being used.

Key Takeaway

Feature selection is about identifying the **most useful set of variables** for a machine learning problem rather than simply reducing the number of columns.

The three major approaches can be remembered as:

Filter → Statistical evaluation

Wrapper → Model-based evaluation

Embedded → Selection during model training

The overall goal is to remove unnecessary information while retaining the features that provide meaningful value to the machine learning model.